

Lecture 5 -Instrument Variables

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- Instrument Variables
- Application: Returns to Education
 - Omitted Variable Bias - “ability bias”
 - Selection Bias

Population Regression vs. Causal Model

- Population Regression Model (PRM):

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i = x_i' \beta + u_i$$

in which $\beta = \operatorname{argmin}_b E[(y_i - x_i' b)^2]$ (1)

- PRM: β is defined by f.o.c.: $E[x_i(y_i - x_i' \beta)] = 0$. That is, the residual $u_i := y_i - x_i' \beta$ satisfies $E[x_i u_i] = 0$. We often write PRM as $E^*[y_i | x_i] = x_i' \beta$.
- OLS: $\hat{\beta}$ is sample analogue of PRM, defined by

$$\frac{1}{N} \sum_{i=1}^N x_i (y_i - x_i' \hat{\beta}) = 0$$

Let's of good things about PRM/OLS - $E^*[y|x]$ as best linear approximation to $E[y_i|x_i]$, Frish-Waugh, omitted variables formula,...

Population Regression vs. Causal Model

- Causal model: Imagine we are thinking about a manipulation of x_{1i} . We ask “what is the **causal effect** of a change in x_{1i} , holding constant all other factors?” To address this, we assume for simplicity

$$y_i = \beta_0 + \beta_1 x_{1i} + u_i \quad (2)$$

i.e., the causal effect of x_1 is linear, with additive effect of all other factors. We can assume $E[u_i] = 0$, but ***we cannot be sure*** that $u_i \perp x_{1i}$. We may well have:

$$E[x_{1i} u_i] \neq 0 \quad (3)$$

From OVT, population regression coefficient is

$$\begin{aligned} \beta_1^{PRM} &= \frac{E[(x_{1i} - E[x_{1i}]) y_i]}{E[(x_{1i} - E[x_{1i}])^2]} \\ &= \beta_1 + \lambda \\ \lambda &= E[(x_{1i} - E[x_{1i}])^2]^{-1} E[(x_{1i} - E[x_{1i}]) u_i] \end{aligned} \quad (4)$$

where $\lambda \neq 0$ if $E[x_{1i} u_i] \neq 0$. You may interpret it as omitted variable bias: $u_i = \alpha_i + \nu_i$, $E[x_i \nu_i] = 0$ but $E[x_i \alpha_i] \neq 0$, α_i is omitted from the regression.

Population Regression vs. Causal Model

- To see this in OLS (sample analogue of PRM):

$$\begin{aligned}\hat{\beta}_1^{OLS} &= \left(\frac{1}{N} \sum_i x_i x_i' \right)^{-1} \frac{1}{N} \sum_i x_i y_i \\ &= \left(\frac{1}{N} \sum_i x_i x_i' \right)^{-1} \frac{1}{N} \sum_i x_i (x_i \beta_1 + u_i) \\ \Rightarrow \hat{\beta}_1^{OLS} - \beta_1 &= \left(\frac{1}{N} \sum_i x_i x_i' \right)^{-1} \frac{1}{N} \sum_i x_i u_i \xrightarrow{p} 0\end{aligned}$$

$\hat{\beta}_1^{OLS}$ is a consistent estimate for $\beta_1^{PRM} = \beta_1 + \lambda$, which is not β_1 in the causal model whenever $E[x_i u_i] \neq 0$.

- Let's now consider the method instrumental variables (IV) also called **two-stage least squares (2sls)**. Assume there is some set of variables z_i with $\#rows(z) \geq \#rows(x)$ and

$$\begin{aligned} E[z_i u_i] &= 0 \\ cov(z_i, x_i) &\neq 0 \end{aligned} \tag{5}$$

Why is z_i this useful?

- we find the part of x_i that is explained by z_i , denoted by $\hat{x}$, and use only that in estimating the regression.
- there is very importantly an exclusion restriction that $E[z_i u_i] = 0$, which implies $E[\hat{x}_i u_i] = 0$, allowing us to retrieve a consistent estimator for β via OLS.
- This is implemented by regressing each row of x on z , then using the predicted fits ($\hat{x}'s$) as explanatory variables in a model for y . Hence the name “two stage” least squares.

Instrument Variables

- Let's start with the causal model with one regressor and a constant: $(1, x_i)$:

$$y_i = \beta_0 + \beta_1 x_i + u_i, E[u_i x_i] \neq 0$$

- We assume that there is some z_i such that: (a) $E[z_i u_i] = 0$ and (b) $\text{cov}[x_i, z_i] \neq 0$. We first get the part of x that is “clean” by the “first stage” regression:

$$x_i = \underbrace{\pi_0 + \pi_1 z_i}_{\text{pred}x_i} + \eta_i \quad (6)$$

- When we run this regression in sample, we know the FOC will require

$$\frac{1}{N} \sum_i z_i (x_i - \text{pred}x_i) = \frac{1}{N} \sum_i z_i (x_i - (\hat{\pi}_0 + \hat{\pi}_1 z_i)) = 0$$

where the predicted/fitted x from the sample OLS:

$$\text{pred}x_i = \hat{\pi}_0 + \hat{\pi}_1 z_i.$$

Instrument Variables

- Now we consider a regression of y_i on a constant and $\hat{pred}x_i$. The OLS coefficient from this “second stage” regression will be:

$$\hat{\beta}_1^{iv} = \frac{\frac{1}{N} \sum_i (\hat{pred}x_i - \overline{\hat{pred}x_i}) y_i}{\frac{1}{N} \sum_i (\hat{pred}x_i - \overline{\hat{pred}x_i})^2}$$

and now let's write:

$$y_i = \beta_0 + \beta_1 x_i + u_i = \beta_0 + \beta_1 \hat{pred}x_i + \underbrace{u_i + \beta_1(x_i - \hat{pred}x_i)}_{\text{orthogonal to } \hat{pred}x_i!}$$

note $u_i + \beta_1(x_i - \hat{pred}x_i)$ is uncorrelated (orthogonal) with $(1, \hat{pred}x_i)$, which means we can get a consistent estimate of β_1 - the causal effect of x in the causal model!

- To obtain a consistent estimate of β_1 in the model

$$y_i = \beta_0 + \beta_1 x_i + u_i, \quad E[u_i] = 0 \text{ but } E[x_i u_i] \neq 0$$

- Find instrument z that satisfies (1) relevance $\text{cov}(z_i, x_i) \neq 0$, and (2) exclusion restriction $\text{cov}(z_i, u_i) = 0$.
- ① Estimate the first stage:

$$x_i = \pi_0 + \pi_1 z_i + \eta_i$$

form estimates of $\text{pred}x_i = \pi_0 + \pi_1 z_i$.

- ② Estimate the 2nd stage: replace x_i by $\text{pred}x_i$:

$$y_i = \beta_0 + \beta_1 \text{pred}x_i + (u_i + \beta_1(x_i - \text{pred}x_i))$$

which satisfies $E[\text{pred}x_i \times (u_i + \beta_1(x_i - \text{pred}x_i))] = 0$.

- Let's look more closely at the (population) IV formula with 1 regressor:

$$\beta_1^{iv} = \frac{E[(predx_i - E[predx_i]) \times y_i]}{E[(predx_i - E[predx_i])^2]} \quad (7)$$

$$predx_i = \pi_0 + \pi_1 z_i \text{ (first stage)}$$

$$\Rightarrow predx_i - E[predx_i] = \pi_1(z_i - E[z_i])$$

$$\begin{aligned} \Rightarrow \beta_1^{iv} &= \frac{E[\pi_1(z_i - E[z_i])y_i]}{E[\pi_1^2(z_i - E[z_i])^2]} \\ &= \frac{1}{\pi_1} \frac{cov(z_i, y_i)}{var(z_i)} = \frac{\delta_1}{\pi_1} \end{aligned}$$

where δ_1 is the slope coefficient from the population reg.:

$$y_i = \delta_0 + \delta_1 z_i + \nu_i, \quad E[\nu_i] = E[\nu_i z_i] = 0$$

$$\rightarrow \delta_1 = \frac{cov(y_i, z_i)}{var(z_i)}$$

β^{iv} vs. Reduced Form

- Focus on $\beta_1^{iv} = \frac{\delta_1}{\pi_1}$:

$$\begin{aligned}y_i &= \beta_0 + \beta_1 x_i + u_i && \text{(structural model)} \\x_i &= \pi_0 + \pi_1 z_i + \eta_i && \text{(1st stage)} \\ \Rightarrow y_i &= (\beta_0 + \beta_1 \pi_0) + (\beta_1 \pi_1) z_i + (u_i + \beta_1 \eta_i) \\ &= \delta_0 + \delta_1 z_i + \nu_i && \text{(reduced form)}\end{aligned} \tag{8}$$

where

$$\begin{aligned}\delta_0 &= \beta_0 + \beta_1 \pi_0 \\ \delta_1 &= \beta_1 \pi_1 \\ \nu_i &= u_i + \beta_1 \eta_i\end{aligned}$$

- The “structural equation” $y_i = \beta_0 + \beta_1 x_i + u_i$ and the “first stage model” imply a relationship between z and y :

$$\begin{aligned}y_i &= (\beta_0 + \beta_1 \pi_0) + \beta_1 \pi_1 z_i + (u_i + \beta_1 \eta_i) \\ &= \delta_0 + \delta_1 z_i + \nu_i\end{aligned}$$

This is called the “reduced form” (RF) model.

- Notice that the reduced form error $\nu_i = u_i + \beta_1 \eta_i$ is “OK”, since by exclusion restriction, and first-stage model,

$$E[z_i \nu_i] = E[z_i u_i] + \beta_1 E[z_i \eta_i] = 0.$$

So the population r.f. coefficient will be $\delta_1 = \beta_1 \pi_1$. We “unpack” β_1 from the reduced form estimate of $\beta_1 \pi_1$ by dividing by our estimate of π_1 .

- **Sample analogue:** All the same arguments work for sample 1st stage and sample RF. We estimate the RF by OLS and get an estimate of $\hat{\delta}_1 = \widehat{\beta_1 \pi_1}$. And so to estimate β_1 we form:

$$\hat{\beta}_1 = \frac{\hat{\delta}_1}{\hat{\pi}_1}$$

Multiple Instruments

- So far, we have been assuming there is only one x in the structural model. However, the same process works when we have a more complex structural model with other covariates. Consider:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \dots + \beta_K x_{Ki} + u_i$$

Let's assume that the "other regressors" $x_{0i} = (x_{2i}, x_{3i} \dots x_{Ki})'$ are uncorrelated with u_i but regressor 1 is problematic: $E[x_{ji} u_i] = 0$ for $j = 2 \dots K$ but $E[x_{1i} u_i] \neq 0$.

- Think of an example like Mincer regression:

$$y_i = \beta_0 + \beta_1 \text{Educ}_i + \beta_2 \text{age}_i + u_i$$

We aren't worried that unobserved ability is correlated with age (older but not wiser), but we are worried about schooling.

- We follow the same process. We have z_i s.t. $E[z_i u_i] = 0$ and z_i helps predict x_{1i} , conditional on the other regressors.

Multiple Instruments

- 1 We fit the “first stage” regression:

$$x_i = \pi_0 + \pi_1 z_i + x'_{Oi} \pi_O + \eta_i$$

form $\text{pred}x_{1i} = \pi_0 + \pi_1 z_i + x'_{Oi} \pi_O$

- 2 Fit the “second stage” regression of y on $\text{pred}x_{1i}$ and x_{Oi} :

$$y_i = \beta_0 + \beta_1 \text{pred}x_{1i} + x'_{Oi} \beta_O + e_i$$

Note that we can think of this as predicting $(x_{1i}, x_{2i}, x_{3i} \dots x_{Ki})$ using $(z_i, x_{2i}, x_{3i} \dots x_{Ki})$ and fitting a second stage model of y on all the predicted regressors.

- Note: we need to **include** x_{Oi} **in both the 1st stage and the 2nd stage!!!** Think of x_{Oi} as their own instruments. The model is under-identified ($\#z\text{'s} < \#x\text{'s}$) if x_{Oi} are not included in the first stage.

Multiple Instruments

- Why does this work? As in the 1-variable case:

$$y_i = \beta_0 + \beta_1 x_{1i} + x'_{0i} \beta_0 + u_i \quad (\text{structural model})$$

$$x_i = \pi_0 + \pi_1 z_i + x'_{0i} \pi_0 + \eta_i \quad (\text{1st stage})$$

$$\begin{aligned} \Rightarrow y_i &= \beta_0 + \beta_1 \pi_0 + (\beta_1 \pi_1) z_i + x'_{0i} (\beta_0 + \beta_1 \pi_0) + u_i + \beta_1 \eta_i \\ &= \delta_0 + \delta_1 z_i + x'_{0i} \delta_0 + \nu_i \end{aligned}$$

So if we estimate the reduced form model relating y to z and x_0 , the coefficient on z_i is $\delta_1 = \beta_1 \pi_1$. We can unpack the estimate of β_1 by dividing by the coefficient of z_i in the first stage model, π_1 . And as in the 1-variable case this all works for the sample regressions too. We get $\hat{\delta}_1 = \widehat{\beta_1 \pi_1}$ and we divide by $\hat{\pi}_1$.

- Using Frisch-Waugh: define $\tilde{z}_i = z_i - E^*[z_i|1, x_{oi}]$,

$$\delta_1 = \frac{\text{cov}(y_i, z_i - E^*[z_i|1, x_{oi}])}{\text{var}(\tilde{z}_i)}$$

$$\pi_1 = \frac{\text{cov}(x_i, \tilde{z}_i)}{\text{var}(\tilde{z}_i)}$$

$$\rightarrow \beta_1^{IV} = \frac{\text{cov}(y_i, \tilde{z}_i)}{\text{cov}(x_i, \tilde{z}_i)}$$

Exercise: apply Frisch-Waugh to OLS (1st stage, and 2nd stage)
and prove $\hat{\beta}_1^{IV} = \frac{\hat{\delta}_1}{\hat{\pi}_1}$.

Asymptotic Inference of 2SLS

- Back to 2SLS:

$$y_i = \beta_0 + \beta_1 x_{1i} + x'_{0i} \beta_0 + u_i \quad (\text{structural model})$$

$$x_i = \pi_0 + \pi_1 z_i + x'_{0i} \pi_0 + \eta_i \quad (\text{1st stage})$$

$$y_i = \beta_0 + \beta_1 \text{pred}x_i + x'_{0i} \beta_0 + (u_i + \beta_1 \eta_i) \quad (\text{2nd stage})$$

Let Γ represent the coefficients in the 2nd stage, and $V'_i = (1, \text{pred}x_i, x'_{0i})$, OLS estimates:

$$\begin{aligned}\hat{\Gamma} &= \left(\sum_i V_i V'_i \right)^{-1} \sum_i V_i y_i \\ &= \left(\sum_i V_i V'_i \right)^{-1} \sum_i V_i (V'_i \Gamma + u_i + \beta_1 \eta_i) \\ &= \Gamma + \left(\sum_i V_i V'_i \right)^{-1} \sum_i V_i (u_i + \beta_1 \eta_i)\end{aligned}$$

- Apply what we prove in the first week: assume homoscedasticity,

$$\sqrt{n}(\hat{\Gamma} - \Gamma) \rightarrow^d N(0, E[V_i V'_i]^{-1} \times \text{var}(u_i + \beta_1 \eta_i))$$

Asymptotic Inference of 2SLS

- Note: a common mistake is to estimate the first stage and the 2nd stage separately instead of as a system of equations. Why does it matter? The estimates would be the same, but **the standard errors are not!** Stare at the asymptotic distribution for a few seconds.

$$\begin{aligned} u_i &= y_i - (\beta_0 + \beta_1 x_{1i} + x'_{0i} \beta_0) \text{ residual in causal model} \\ &\neq y_i - (\beta_0 + \beta_1 \text{pred}x_{1i} + x'_{0i} \beta_0) \text{ residual in 2nd stage} \end{aligned}$$

if the 2nd stage is estimated as a regular OLS, the standard errors for $\hat{\Gamma}$ will be based on the variance of

$\hat{u}_i = y_i - (\beta_0 + \beta_1 \text{pred}x_{1i} + x'_{0i} \beta_0)$, which is not u_i .

$$\sqrt{n}(\hat{\Gamma} - \Gamma) \rightarrow^d N(0, E[V_i V_i']^{-1} \times \text{var}(\mathbf{u}_i + \beta_1 \eta_i))$$

Returns to Education

- A convincing analysis of the causal link between education and earnings require an exogenous source of variation in education outcomes.
- Start from population regression (a la Mincer):

$$y_i = \beta_0 + \beta_1 S_i + \beta'_x x_i + u_i$$

$$E[u_i] = E[u_i x_i] = 0,$$

$$E[u_i S_i] \neq 0$$

y_i =log wage of person i , S_i =yrs. of schooling, x_i =experience (and maybe other covariates).

- $\hat{\beta}_1^{OLS}$ often ranges from 5-15 percent: earnings are 5-15% higher on average when a person gets an additional year of schooling.

Returns to Education

OLS estimates are biased as long as $E[u_i S_i] \neq 0$.

- ① Ability bias (Griliches 1977): $u_i = \beta_a \times Ability_i + \epsilon_i$, with $E[Ability_i \times S_i] \neq 0$ while $E[\epsilon_i S_i] = 0$. With auxiliary regression

$$Ability_i = \pi_0 + \pi_1 S_i + \pi'_x x_i + \xi_i$$

OLS of y_i on $(1, S_i, x_i)$ would yield

$$\hat{\beta}_1^{OLS} \rightarrow^p \beta_1 + \beta_a \pi_1 > \beta_1 \text{ if } \beta_a \pi_1 > 0$$

- ② Heterogeneous returns to education: suppose the returns to education $\{\beta_i\}$ vary across individuals, and higher- β_i individuals obtain more years of schooling, the OLS estimate could again be upward-biased, driven by high- β_i individuals (see Card 2001)

- ③ The estimate can also be downward biased due to **measurement errors**. Suppose in the data we observe $S_i = S_i^* + \nu_i$, where S_i^* is the true education while ν_i is the measurement error that satisfies $\nu_i \perp (1, S_i^*, x_i, u_i)$. So $y_i = \beta_0 + \beta_1 S_i^* + \beta'_x x_i + (\beta_1 \nu_i + u_i)$

$$\hat{\beta}_1^{OLS} = \frac{\text{cov}(y_i, S_i - E^*[S_i|x_i])}{\text{var}(S_i - E^*[S_i|x_i])} = \frac{\text{cov}(y_i, S_i^* - E^*[S_i^*|x_i])}{\text{var}(S_i^* - E^*[S_i^*|x_i]) + \text{var}(\nu_i)}$$

Even if $E[u_i S_i] = 0$, we still get a downward-biased estimate due to measurement error when $\text{var}(\nu_i) > 0$. This is called **attenuation bias**.

Instrument for Education

- Card (1993) explores the use of proximity to college as an exogenous determinant of schooling. $z_i = 1$ if i lived near a 4-year college when teenager.

- Model:

$$y_i = \beta_0 + \beta_1 S_i + \beta'_x x_i + u_i$$

y_i = log wage measured in 1976 (age 26-34). x_i includes regional and family background factors such as parental education and family structure.

- 1st stage:

$$S_i = \pi_0 + \pi_1 z_i + \pi'_x x_i + \eta_i$$

$\hat{\pi}_1 > 0$: men who grew up in local labor markets with a nearby college have significantly higher education.

- Reduced form:

$$y_i = \delta_0 + \delta_1 z_i + \delta'_x x_i + \epsilon_i$$

- IV model (2nd stage):

$$y_i = \beta_0 + \beta_1 \text{pred} S_i + \beta'_x x_i + (u_i + \beta_1 \eta_i)$$

Instrument for Education

First Stage, Reduced Form, and Structural Models for Return to Education

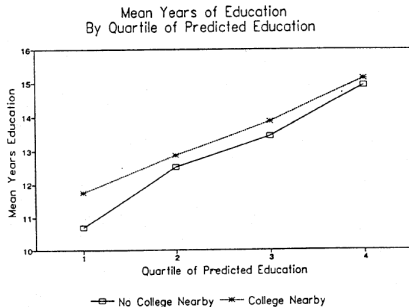
	OLS Model	First Stage Model	Reduced Form	IV Model
Live Near 4-year College as Youth (1966)	--	0.322 (0.083)	0.045 (0.018)	--
Years of Education	0.073 (0.004)	--	--	0.140 (0.055)
Other Controls	YES	YES	YES	YES
Sample Size	3,010	3,010	3,010	3,010

note $\hat{\beta}_1^{IV} = \frac{\hat{\delta}_1}{\hat{\pi}_1} \gg \hat{\beta}^{OLS}$ OLS estimates appear downward biased rel. to IV.

- Could be attenuation bias from measurement errors in S_i . But researchers estimate this may only explain 10-15% of the diff.

Instrument for Education

- IV estimates could be larger than true β_1 if it's driven by lower-income families who have higher returns to education (they face higher borrowing costs and kids might stop schooling too soon).
- It does appear like the instrument - proximity to college matters more to lower-income households.



Note: prediction equation is fit to subsample with no college nearby

Instrument for Education

- The higher effect of a nearby college on schooling is also exploited to test **the exclusion restriction** that z_i does not have a **direct** effect on earnings (only indirectly through S_i).
- Specifically, include z_i itself as a control, and use $z_i \times P_i$, the interaction between proximity to college and family background (indicator for low family income), as an instrument for S_i : this identification strategy relies on the extra boost to education among children with poor family backgrounds.

$$S_i = \pi_0 + \pi_z z_i + \pi_x x' + \underbrace{\pi_2 (z_i \times P_i)}_{\text{instrument}} + \eta_i$$
$$y_i = \beta_0 + \underbrace{\beta_z z_i}_{\text{control}} + \beta_x x' + \beta_1 S_i + \xi_i$$

- (First stage $z \times P$ as instrument for S): the extra boost to education among children with poor family backgrounds.
- (Exclusion restriction of $z \times P$): the direct earnings effects of living near a college $\beta_z z_i$ do not vary by family background P_i .

Instrument for Education

Table 5: Instrumental Variables Estimates of the Return to Education
Based on Interaction of Parental Education and Proximity to College

	Reduced Form Models:		Structural Models	
	Education	Earnings	of Earnings	
	(1)	(2)	(3)	(4)
1. Live Near College in 1966	0.154 (0.135)	0.029 (0.024)	0.015 (0.029)	0.013 (0.024)
2. Live Near College * Low Parental Education ^a	0.462 (0.186)	0.043 (0.032)	--	--
3. Education ^b	--	--	0.093 (0.065)	0.097 (0.048)

- $\hat{\beta}_1^{iv}$ in this alternative model remains higher than the OLS estimate (0.073).
- If β_z is significantly positive, it suggests living close to a college has a direct effect on schooling, which violates the exclusion restriction.

- Columns (3)-(4) show $\hat{\beta}_z$ is small and insignificantly different from 0. It provides no evidence against the original identifying assumption that college proximity (z) is an exogenous determinant of schooling.

- Compulsory schooling laws
- Lottery
- Month of birth

In each case there are concerns about the exclusion restriction

$E[z_i u_i] = 0$:

- whether z has a direct effect on the outcome
- whether there is selection into z_i based on unobservables that are correlated with u_i

That being said, we have to start somewhere, and there are tests for the exclusion restriction like in the example above. The other exogenous variables x_i also have to be chosen carefully to address alternative explanations.

A Model of Endogenous Schooling

- Regression models of earnings on schooling are derived from (structural) models of endogenous education choice.
- Rational individuals reach an optimal schooling decision by balancing the benefits of higher schooling against the costs. They maximize the discounted present value of earnings, net of tuition & opportunity costs for staying in school rather than working.
- Here we look at a simplified version of the model in Card (2001) - “Estimating the return to schooling...”
 - assume individuals live in infinite horizon and consume a constant amount every year (an extreme form of consumption smoothing) that is allowed by the NPV of lifetime earnings:

$$\begin{aligned} & \max_s \int_0^{\infty} u(c) e^{-\rho t} dt \\ \text{s.t. } & \frac{c}{R} = \underbrace{\int_s^{\infty} y(s, t) e^{-Rt} dt}_{\text{earnings}} - \underbrace{\int_0^s T(t) e^{-Rt} dt}_{\text{tuition \& OC}} \end{aligned}$$

A Model of Endogenous Schooling

in this case it is equivalent to maximize $u(c)$ subject to the budget constraint w consumption smoothing. The FOC for schooling can be written as:

$$\begin{aligned} \frac{d}{ds} \int_s^{\infty} y(s, t) e^{-Rt} dt &= \frac{d}{ds} \int_0^s T(t) e^{-Rt} dt \\ \underbrace{\int_s^{\infty} \frac{\partial y(s, t)}{\partial s} e^{-Rt} dt}_{MB(s) \text{ marginal benefits}} &= \underbrace{(y(s, s) + T(s)) \times e^{-Rs}}_{\text{marginal cost } MC(s)} \end{aligned}$$

- Assume Mincer $y(s, t) = f(s) \times h(t) = f(s) * 1$,

$$\begin{aligned} MB(s) &= \int_s^{\infty} \frac{\partial y(s, t)}{\partial s} e^{-Rt} dt = f'(s) \times \frac{e^{-Rs}}{R} \\ &= MC(s) \\ \Rightarrow \frac{f'(s)}{f(s)} &= \underbrace{R + R \times T(s)/f(s)}_{d(s)} \end{aligned}$$

- The marginal benefits & marginal costs could vary by individuals:
 - lower-income households facing higher interest rates R ;

A Model of Endogenous Schooling

- the disadvantaged groups could benefit more from an additional year in school;
 - some individuals may have a higher disutility of studying (rel. to working).
- *Note: the model above won't be tested, but the econometric framework below is fair for homework and exams.*

Heterogeneous Returns to Schooling

- The model motivates a specification of the heterogeneous MB/MC:

$$\begin{aligned}f'(s)/f(s) &= b_i - k_1 s \\ d(s) &= r_i + k_2 s\end{aligned}$$

where (b_i, r_i) vary by individual! It implies a model for log earnings:

$$\begin{aligned}\log(y_i) &= \alpha_i + b_i S_i - \frac{k_1}{2} S_i^2 \\ &= a_0 + \bar{b} S_i - \frac{k_1}{2} S_i^2 + \underbrace{a_i + (b_i - \bar{b}) S_i}_{\text{residual}}\end{aligned}$$

- If we estimate the model above via an OLS regression of log earnings on $(1, S_i)$, can we retrieve the average $\bar{b}$ return to schooling in the population? No if

$$E[S_i \times (a_i + (b_i - \bar{b}) S_i)] \neq 0$$

Heterogeneous Returns to Schooling

- Auxiliary regressions:

$$a_i = \lambda_0(S_i - \bar{S}) + u_i$$

$$b_i - \bar{b} = \psi_0(S_i - \bar{S}) + \nu_i$$

plugged into the earnings model:

$$\log(y_i) = \text{const} + (\bar{b} + \lambda_0 - \psi_0\bar{S}) S_i + \left(-\frac{k_1}{2} + \psi_0\right) S_i^2 + (u_i + \nu_i S_i)$$

assume $E[\nu_i|S_i] = 0$, show $E[\nu_i S_i^2] = 0$ (lecture1), in which case OLS estimates of $\log(y)$ on $(1, S_i)$ converge to

$$\hat{b}^{OLS} \rightarrow^p \bar{b} + \lambda_0 + \psi_0\bar{S}$$

- λ_0 : ability bias
- $\psi_0\bar{S}$: selection into higher schooling; $\psi_0 > 0$ represents a positive selection into education.

Heterogeneous Returns to Schooling

- What if we have an instrument for S_i ? If high- b_i individuals respond more to the instrument (e.g., nearby college),

$$\beta_1^{IV} = \frac{\text{cov}(y_i, z_i)}{\text{cov}(S_i, z_i)} = \frac{\int_{b_i} \text{cov}(y_i, z_i | b_i) dF(b_i)}{\int_{b_i} \text{cov}(S_i, z_i | b_i) dF(b_i)}$$

will be dominated by the high- b_i individuals – that is we are likely to the average return among a sub-population who responds more to z_i .

- This point would become more clear when we discuss the local average treatment effects by Angrist and Imbens!